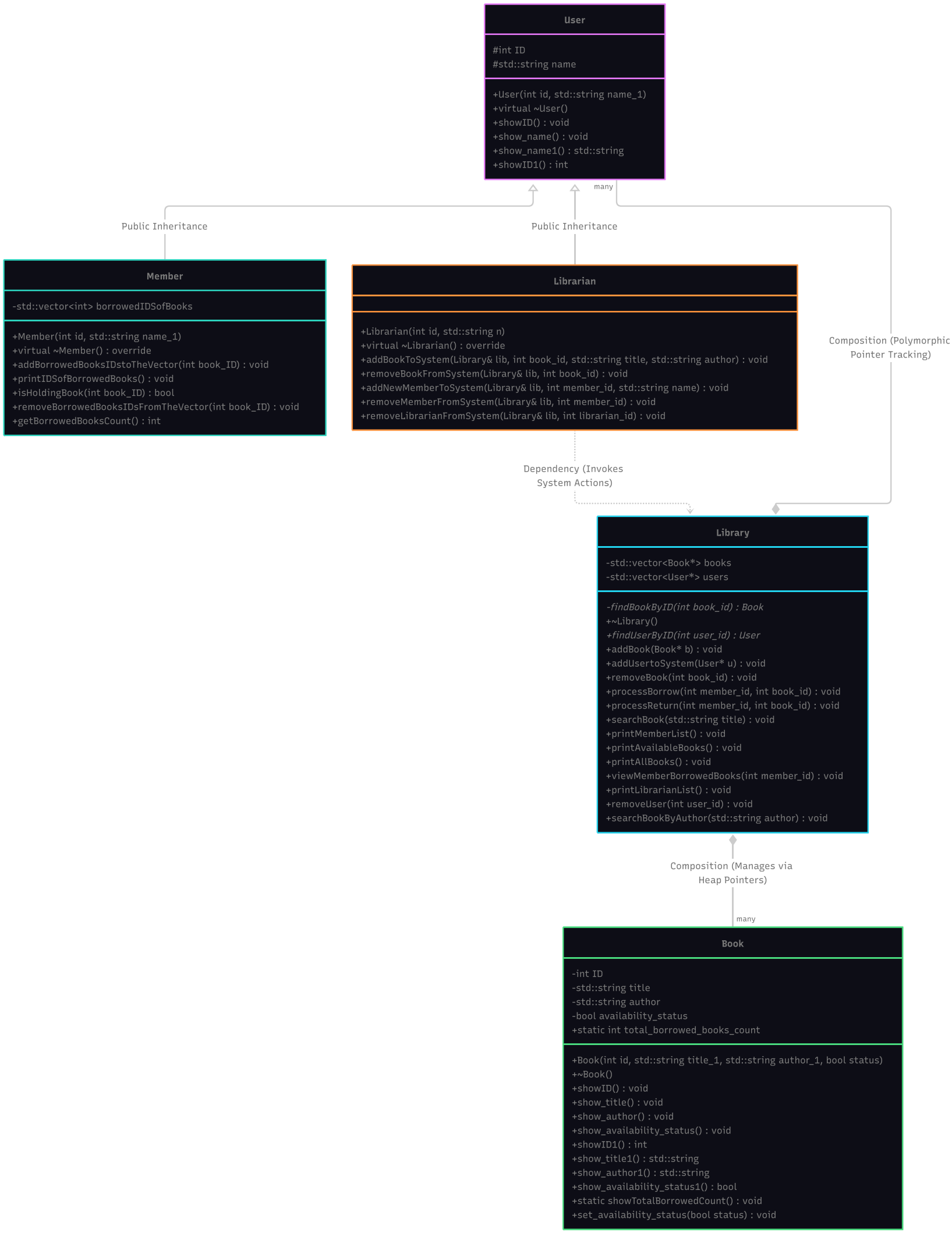

User
#int ID
#std::string name
+User(int id, std::string name_1)
+virtual ~User()
+showID() : void
+show_name() : void
+show_name1() : std::string
+showID1() : int
many
Public Inheritance
Public Inheritance
Member
-std::vector<int> borrowedIDSofBooks
+Member(int id, std::string name_1)
+virtual ~Member() : override
+addBorrowedBooksIDstoTheVector(int book_ID) : void
+printIDSofBorrowedBooks() : void
+isHoldingBook(int book_ID) : bool
+removeBorrowedBooksIDsFromTheVector(int book_ID) : void
+getBorrowedBooksCount() : int
Librarian
+Librarian(int id, std::string n)
+virtual ~Librarian() : override
+addBookToSystem(Library& lib, int book_id, std::string title, std::string author) : void
+removeBookFromSystem(Library& lib, int book_id) : void
+addNewMemberToSystem(Library& lib, int member_id, std::string name) : void
+removeMemberFromSystem(Library& lib, int member_id) : void
+removeLibrarianFromSystem(Library& lib, int librarian_id) : void
Composition (Polymorphic Pointer Tracking)
Dependency (Invokes System Actions)
Library
-std::vector<Book*> books
-std::vector<User*> users
-findBookByID(int book_id) : Book
+~Library()
+findUserByID(int user_id) : User
+addBook(Book* b) : void
+addUsertoSystem(User* u) : void
+removeBook(int book_id) : void
+processBorrow(int member_id, int book_id) : void
+processReturn(int member_id, int book_id) : void
+searchBook(std::string title) : void
+printMemberList() : void
+printAvailableBooks() : void
+printAllBooks() : void
+viewMemberBorrowedBooks(int member_id) : void
+printLibrarianList() : void
+removeUser(int user_id) : void
+searchBookByAuthor(std::string author) : void
Composition (Manages via Heap Pointers)
many
Book
-int ID
-std::string title
-std::string author
-bool availability_status
+static int total_borrowed_books_count
+Book(int id, std::string title_1, std::string author_1, bool status)
+~Book()
+showID() : void
+show_title() : void
+show_author() : void
+show_availability_status() : void
+showID1() : int
+show_title1() : std::string
+show_author1() : std::string
+show_availability_status1() : bool
+static showTotalBorrowedCount() : void
+set_availability_status(bool status) : void